INCH-POUND
MIL-DTL-0060945C (AR)
05 April 2004
USED IN LIEU OF
MIL-C-60945B (AR)
12 May 1993

DETAIL SPECIFICATION

PROJECTILE ASSEMBLY, FOR CARTRIDGE, 40MM: PRACTICE, M385A1

This specification is approved for interim use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC). Other activities in the Department of Defense may use this interim revision or may continue using MIL-C-60945B (AR).

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Projectile Assembly for the Cartridge, 40MM: Practice, M385A1.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-A-8625 - Anodic Coatings for Aluminum and Aluminum Alloys

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, New Jersey 07806-5000, or e-mailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil

AMSC N/A FSC 1310
DISRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation (see 6.2).

U.S. ARMY (ARDEC) DRAWINGS (see 6.6)

8866714 - Projectile Assembly, For Cartridge, 40MM: Practice, M385A1
8866715 - Body
8886402 - Blank, Band

(Copies of these drawings may be requested online at drawing-request@pica.army.mil, or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny NJ 07806-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS INTERNATIONAL (ASTM)

ASTM B117 - Method of Test for Salt Spray (fog) Testing
ASTM D3951 - Standard Practice for Commercial Packaging

(Copies of this document are available online at www.astm.org or from the ASTM International, 100 Barr Harbor Drive, Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the solicitation or contract order (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials, parts and assemblies shall comply with requirements specified on the applicable drawings and referenced specifications.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

3.4 Workmanship. All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. VERIFICATION

TABLE I. Requirement/Inspection Cross Reference Matrix

Method of inspection
N/A - Not Applicable
1 - Analysis
2 - Demonstration
3 - Examination
4 - Test

Classes of inspection
A - First Article
B - Conformance

Section 3 Requirement	Inspection Methods					Inspection Class		Section 4 Inspection Method
	N/A	1	2	3	4	A	B	
3.1			X	X	X	X		4.3
3.2			X	X	X	X	X	4.4, 4.5.1, 4.5.2, 4.5.3
3.3				X		X	X	4.5.4
3.4				X		X	X	4.5.5

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection Condition. Unless otherwise specified, all inspection shall be performed in accordance with the test conditions specified in section 4.5.

4.3 First Article. When specified in the solicitation or contract, a sample of the Projectile Assembly, 40MM: Practice, M385A1 shall be subjected to first article inspection in accordance with Table II and 4.3.1.

4.3.1 First Article Quantity. The contractor shall submit a first article for evaluation in accordance with the provision of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Parts description	Drawings	Quantity
Body	8866715	50
Projectile Assembly	8866714	50

4.3.2 Inspections to be performed. The first article inspection shall be 100 % examination of all drawing characteristics (listed and unlisted) including workmanship requirements for the parts and assemblies listed in 4.3.1 and the tests specified in Table II herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

4.3.3 First article rejection. If any item of the sample fails to comply with the first article requirements, the sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

TABLE II. First Article Inspection

Projectile Assembly for the 40MM Cartridge, Practice M385A1				
	EXAMINATION OR TEST	N0. OF SAMPLE UNITS	REQUIREMENT PARAGRAPH	INSPECTION METHOD
	Body (Dwg. 8866715)			
	Salt Spray	10	3.2	4.5.1
	Projectile Assembly (Dwg. 8866714)			
	Push test of rotating band	50	3.2	4.5.2
Notes:				

4.4 Conformance inspection.

4.4.1 Inspection lot formation. Lot formation shall be in accordance with MIL-STD-1916. For the components identified below, each inspection lot of projectiles shall be produced with components complying with the following constraints:

a. Metal parts or components from one interfix number, from one manufacturer

4.4.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for Critical, Major and Minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.4.2.1 through 4.4.2.2. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.1	TITLE Body	SHEET 1 OF 1		DRAWING NUMBER 8866715
				NEXT HIGHER ASSEMBLY 8866714
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Diameter of rotating band surface (after knurl)	Level IV	3.2	Gage
102.	Location of basic diameter on spherical radius	Level IV	3.2	Gage
103.	Spherical radius	Level IV	3.2	Gage
104.	Location of basic diameter (excluding cut-off nib)	Level IV	3.2	Gage
105.	Run-out of indicated area on taper with body diameter	Level IV	3.2	Gage
106.	Body diameter	Level IV	3.2	Gage
107.	Diameter between knurl	Level IV	3.2	Gage
108.	Width of crest knurl	Level IV	3.2	Gage
109.	Width of rotating band surface	Level IV	3.2	Gage
110.	Length from back of rotating band surface to front end of nose	Level IV	3.2	Gage
111.	Location of bourrelet shoulder	Level IV	3.2	Gage
112.	Diameter of bourrelet	Level IV	3.2	Gage
113.	Diameter below bourrelet	Level IV	3.2	Gage
114.	Salt Spray 1/	2 per 8 hr shift	3.3	4.5.1
MINOR				
201.	Protective coating with bare spot on bottom or forward of rotating band groove	Level II	3.2	Visual
202.	Incorrect color	Level II	3.2	Visual
203.	Evidence of poor workmanship	Level II	3.4	Visual / 4.5.5

NOTES:
1/ The production represented by the 8-hour shift sample shall be rejected if any white corrosion is visible to the unaided eye at normal reading distance in any location on the band surface of the body, but not including the edges, when salt spray tested in accordance with 4.5.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

Conformance Inspection

CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH 4.4.2.2	TITLE Projectile Assembly	SHEET 1 OF 1		DRAWING NUMBER 8866714
				NEXT HIGHER ASSEMBLY 8886326
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
CRITICAL	None defined			
MAJOR				
101.	Profile of rotating band	Level IV	3.2	Gage
102.	Diameter of rotating bands	Level IV	3.2	Gage
103.	Diameter of large groove	Level IV	3.2	Gage
104.	Diameter of small groove	Level IV	3.2	Gage
105.	Distance from nose to back of rotating band	Level IV	3.2	Gage
106.	Concentricity of rotating bands with bourrelet and body diameter	Level IV	3.2	Gage
107.	Total length (excluding cut-off nib)	Level IV	3.2	Gage
108.	Push Test of Rotating Band	80 1/	3.2	4.5.2
MINOR				
201.	Marking missing	Level II	3.2	Visual
202.	Evidence of poor workmanship	Level II	3.4	Visual / 4.5.5
	NOTES: 1/ Accept on zero (0) and reject on one (1) or more failures.			

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

4.5 Methods of inspection.

4.5.1 Salt spray. The specified part shall be subjected to the salt spray test in accordance with ASTM B117. This test shall be conducted for 24 hours, or for the duration specified on the applicable drawing, using a salt solution of 5 percent. This test is a destructive test. Parts so tested shall not be returned to the lot. Sampling and rejection criteria is as specified in the applicable referencing paragraph for each part.

4.5.2 Push test of rotating band on projectile assembly. The projectile assembly shall be placed in an approved fixture and the force specified on the drawing shall be applied. Observation shall be made for the requirement specified on the drawing. This test is a destructive test. Parts so tested shall not be returned to the lot.

NOTE: Deformation of the chamfered leading edge of the rotating band caused by compression in the test fixture is permissible. Movement as specified on Dwg. 8866714 is defined as movement of the back face of the rotating band relative to the body.

4.5.3 Inspection equipment. The inspection equipment required to perform the examination and tests prescribed herein is described in the Inspection Method Column in the tables starting with paragraph 4.4.2.1. Inspection equipment used for acceptance of product shall be capable of repeatable measurements, by various experienced inspection/test personnel, to an accuracy of 10% or better of the total tolerance of the characteristic being inspected.

4.5.4 Ammunition lot numbering. Visually verify that an ammunition lot number has been applied to each lot as described in MIL-STD-1168.

4.5.5 Workmanship. Visually verify that all parts and assemblies meet requirements of paragraph 3.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers the requirements, examinations and tests for the Projectile Assembly for the Cartridge, 40MM: Practice, M385A1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number (w/revision level) and date of all reference documents.
c. ASSIST Online to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see Section 2).
d. Requirements for submission of first article sample.
e. Requirement for submission of design verification sample, if applicable.
f. Requirement for submission of inspection equipment designs.
g. Applicable national stock number.
h. Packaging requirements, if other than specified in Section 5.
i. Serialization requirements, if applicable.
j. Certificate of conformance for each lot or shipment of product, if applicable.
k. Government-Furnished Material or equipment, if applicable.
l. Provisions for critical characteristics controls.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs as required to Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000. Request letter of submittal should state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph. Inspection equipment, inspection method and the frequency of inspection should be submitted for approval to ARDEC, ATTN: AMSRD-AAR-QEM-D, Picatinny, NJ 07806-5000.

6.4 Contractor acceptance inspection equipment (AIE). Provision concerning the contractor's AIE used to verify the requirements of this specification should be specified in the contract.

6.5 Ammunition lot numbers. Ammunition requires ammunition data cards in accordance with MIL-STD-1168.

6.6 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development, and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research, and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.7 Definitions.

6.7.1 Critical defect. A critical defect is a defect that judgment and experience indicate is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product; or a defect that judgment and experience indicate is likely to prevent performance of the tactical function of a major end item such as a ship, aircraft, tank, missile, or space vehicle.

6.7.2 Major defect. A major defect is a defect, other than critical, that is likely to result in failure or to reduce materially the usability of the unit of product for its intended purpose.

6.7.3 Minor defect. A minor defect is a defect that is not likely to reduce materially the usability of the unit of product for its intended purpose or is a departure from established standards having little bearing on the effective use or operation of the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.

6.8 Subject term (key word) listing.

Ammunition, 40 MM

6.9 Submission of alternative conformance provisions. All contractor proposed alternative conformance provisions should be submitted to the Government for evaluation/approval as directed by the contracting activity.

6.10 Revisions /Amendments History. The following is a record of changes incorporated into this document:

ECP	Description	Date Approved
R9Q2013	Change to replace variable testing	07/07/1999
R9Q2030	Change to correct mislocated major defects	09/13/1999
R04Q2012	Initial release of M385A1 Projectile Assembly MIL-DTL Specification	Aug 2004

6.11 Interim Revision This interim revision has been prepared at Program Manager request to document in ASSIST Online the version of a draft specification that was placed on contract in 2004. This interim revision has been dated to match the date of the draft specification that was placed on contract. It has been modified slightly to comply with MIL-STD-961 requirements for format, but not content. A fully coordinated revision, compliant with both MIL-STD-961 format and content requirements, will be approved for publication.

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:

Army-AR

(Project 1310-2006-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:07Z
Check the source to verify that this is the current version before use.